



Course Number: 064

Course Title: MakerSpace

Classroom: LB6

Teacher: Ms. Ma. Elena Montes

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Grade Level: 10th - 12th (Elective)

Credits: 0.5 Credits

School Year: 2026-2027

Course Description:

MakerSpace is a hands-on, project-based course that develops creativity, innovation, problem-solving, and technical skills through the design and construction of original projects.

Students will explore the maker design process while working with digital fabrication, electronics, programming, 3D design, and prototyping tools. Activities may include Arduino, CAD and 3D modeling, 3D printing, and introductory Java programming.

Emphasis is placed on planning, experimentation, iteration, documentation, safe use of tools and equipment, collaboration, and real-world applications. Students will learn to evaluate their designs, troubleshoot problems, and improve projects through testing and reflection.

Materials:

Each student is required to bring earphones.

Grading Criteria

Quarter Grading

Formative 40%

Summative 60%

Pass or Fail:

Students will pass or fail the class based on the final grade. According to the current SEP policy reflected in this syllabus, a final grade of 59.5% or above is considered passing.

Classroom Rules and Procedures:

1. The teacher reserves the right to choose your seat for the benefit of the group.
2. You must be seated and ready to start ON TIME as the second entrance bell rings. You have a 5 minute tolerance for coming to class before it is considered TARDY.

Classroom Rules and Procedures:

3. The number of absences permitted for each semester before credit is lost is listed below:
 - 7 total absences
 - 3 tardies = one unexcused absence
 - More than 10 minutes late to class without authorization equals an absence
4. You are expected to complete work within the time established by each assignment. Work submitted 1 day late will receive a maximum grade of 70%.
5. If you are absent, it is your responsibility to check Google Classroom and complete the assigned work.
6. You may leave the classroom ONLY with teacher permission.
7. Eating, drinking, or chewing gum in class is not permitted at any time.
8. Cheating, plagiarism, and unauthorized use of AI tools will result in a ZERO for the assignment in question. AI tools may only be used when explicitly permitted by the teacher. The teacher is the final arbiter in deciding whether academic dishonesty has occurred.
9. Cell phones and personal electronic devices are not permitted during class unless specifically authorized for an activity. Failure to follow this policy may result in the device being confiscated at teacher discretion.
10. Tools, equipment, electronics, 3D printers, and materials may only be used as instructed. Students must follow all safety procedures and handle MakerSpace equipment responsibly. Improper or unsafe use may result in loss of equipment privileges.
11. You have the right to receive instruction in the best possible atmosphere. Any behavior that interferes with this learning environment is inappropriate.

Course Content

Unit 1 - MakerSpace Safety, Design Process and Prototyping

Unit 2 - Introduction to Arduino and Electronics

Unit 3 - Arduino Projects and Physical Computing

Unit 4 - Introduction to 3D Design

Unit 5 - CAD with Fusion 360

Unit 6 - 3D Printing and Slicing

Unit 7 - 3D Modeling with Blender

Unit 8 - Introduction to Java Programming

Unit 9 - Java Game Project (Tetris)

Unit 10 - Independent Maker Project

Unit 11 - Applied AI: Tools, Creation and Digital Publishing